

10. A refrigerant is a chemical used in cooling processes. Some refrigerant compounds can damage the ozone layer.

- (a) The ozone depletion potential (ODP) of a refrigerant compound is the relative amount of damage that it can cause to the ozone layer. The higher the number, the greater the damage.

|   | Refrigerant compound              | Ozone depletion potential |
|---|-----------------------------------|---------------------------|
| 1 | $\text{C}_2\text{F}_4\text{Br}_2$ | 6.00                      |
| 2 | $\text{CF}_2\text{ClBr}$          | 3.00                      |
| 3 | $\text{C}_2\text{FCl}_5$          | 1.00                      |
| 4 | $\text{C}_2\text{F}_3\text{Cl}_3$ | 0.85                      |
| 5 | $\text{C}_2\text{F}_4\text{Cl}_2$ | 0.58                      |
| 6 | $\text{C}_2\text{H}_3\text{Cl}_3$ | 0.16                      |

- (i) Describe a relationship between the formulae of refrigerant compounds 3, 4 and 5 and their ODP.

1

- (ii) Identify which pair of compounds should be used to show the effect of replacing chlorine atoms with bromine atoms in refrigerant compounds.

1

- (iii) The refrigerants carbon dioxide,  $\text{CO}_2$ , and ammonia,  $\text{NH}_3$ , have ODP values of 0.00.

Suggest why this is the case.

1



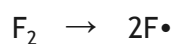
10. (continued)

- (b) The compound difluoromethane,  $\text{CH}_2\text{F}_2$ , is also used as a refrigerant. It is made by reacting fluorine gas with fluoromethane,  $\text{CH}_3\text{F}$ , in a free radical chain reaction.

(i) State what is meant by a free radical.

1

- (ii) The first step in the reaction involves splitting a fluorine molecule to produce two fluorine radicals.



(A) State the name given to this step.

1

(B) Write an equation for a possible propagation step in this reaction.

1

- (c) Household fridges use coolants made from refrigerant compounds. A common coolant is made from 50% difluoromethane,  $\text{CH}_2\text{F}_2$ , ( $GFM = 52 \text{ g}$ ) and 50% pentafluoroethane,  $\text{CF}_3\text{CHF}_2$ , ( $GFM = 120 \text{ g}$ ).

A typical fridge contains 0.05 kg of coolant.

Calculate the number of moles of pentafluoroethane required to make this mass of coolant.

1

[Turn over

